

ASSIGNMENT-18

Problem Statement

A hospital wants to predict whether a patient is **Diabetic (1)** or **Non-Diabetic (0)** based on medical records.

Use the **Diabetes Dataset** from **the dataset provided** and build a KNN and Logistic Regression Classification model.

Tasks

1. Import the required libraries.
2. Load the Diabetes Classification dataset.
3. Convert the dataset into a Pandas DataFrame.
4. Display the first 5 rows.
5. Check the dataset information (`info()`).
6. Check for missing values.
7. Split the dataset into training and testing sets (80:20).
8. Apply **StandardScaler** to scale the features.
9. Train a **K-Nearest Neighbors (KNN)** model with `k = 5`.
10. Train a Logistic Regression Model
11. Predict the test data.
12. Predict the test data using both models.
13. Write **3 observations** based on your results.
14. Identify which model performed better and justify your answer.
15. Write **five observations** based on your analysis.
16. Write a short conclusion.
17. Predict for a New Patient

A new patient visits the hospital with the following medical details:

Feature	Value
Age	45
Glucose	165
BloodPressure	82
BMI	31.5
Insulin	180
DiabetesPedigree	0.85